Work-energy lab

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Compare measured work with kinetic-energy change.

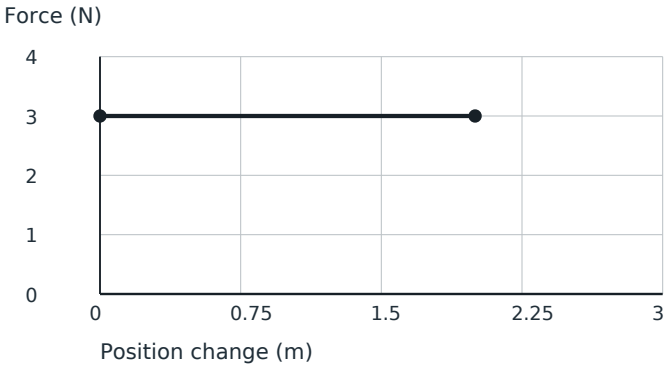
Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 3.*
-15 J; the force opposes displacement.

02. Worked example: explain the model
Work 12 J versus energy change 11 J; check force calibration, distance, speed or omitted friction.

03. Read and interpret the model
6 J.

Model / diagram check



04. Apply the physics
Initial K = 9 J; final K = 4 J; ΔK = -5 J. This does not agree with reported +3 J net work. Check force sign, whether all forces are included, speed measurements and matching endpoints; do not alter measurements to force agreement.

05. Explain your reasoning
Units, force calibration, friction, velocity interval, moving pulley or other energy; preserve actual data.

06. Review: connect an earlier idea *Review: Unit 3, Day 8.*
0 m/s; equal opposite momenta cancel, and the combined mass is 8 kg.

07. Exit check
The energy theorem compares the same interval; mismatched intervals need not agree.

08. STRETCH: test the model
1±2 contains zero; 5±2 does not. Neither result alone identifies the cause.

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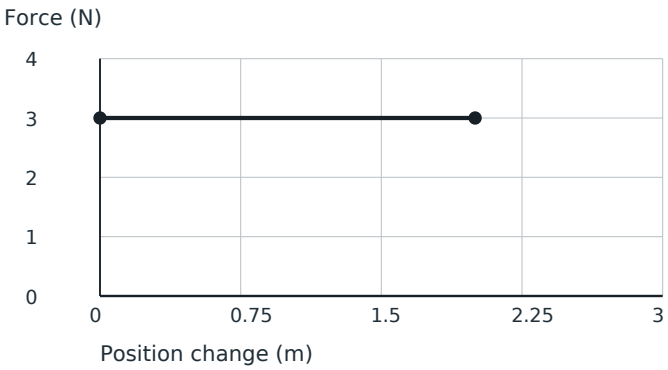
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01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 3.*
W=-15 J because cos 120°=-1/2; the component opposes displacement.

02. Worked example: explain the model
Work 12 J versus energy change 11 J; check force calibration, distance, speed or omitted friction.

03. Read and interpret the model
6 J.

Model / diagram check



04. Apply the physics
ΔK=0.25(1.44-0.16)=0.32 J; resistance work≈-0.03 J.

05. Explain your reasoning
Units, force calibration, friction, velocity interval, moving pulley or other energy; preserve actual data.

06. Review: connect an earlier idea *Review: Unit 3, Day 8.*
v_f=6 m/s. Initial K=720 J; final K=72 J; fraction 0.10 or 10%.

07. Exit check
Work 2 J; speed 2 m/s.

08. STRETCH: test the model
ΔK=8 J; friction work -2 J; mean friction magnitude 1 N.